ENGINEERING RESEARCH CORPORATION LLC

Local, Edge, and Cloud AI: A Workload-Based Decision Guide

Where should your AI workloads run? Measured evidence on performance, privacy, reliability, energy, cost, and administrative effort across local workstations, enterprise servers, edge devices, hosted APIs, and managed cloud platforms — organized by buyer scenario, not by hardware brand.

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This edition audits independently published, manually verified measurements of AI deployment trade-offs and preregisters the ERC workload-based benchmark; it does not present ERC fieldwork results. Edition 2 will report measured results across five deployment models, seven workloads, and ten dimensions, with a machine-readable results file and calculation workbook on the canonical report page listed in the document-control section.

Document control and disclosures

Document control

Table i. Document-control record

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Disclosures

Sponsor or client: None. ERC-initiated and self-funded; future sponsorship will be disclosed on every distributed copy.

Conflicts of interest: ERC does not sell AI hardware, hosting, or inference services and holds no equity in any vendor named. ERC will not accept payment conditioned on findings.

Loans and sponsored equipment (standing commitment): Any loaned, discounted, or vendor-provided hardware or cloud credits used in the ERC benchmark will be labeled in the results file and report, with provider and terms named. Independently purchased equipment is marked as such. Manufacturer specifications are always distinguished from independently measured results.

Editorial independence: Findings are reported regardless of commercial effect on any named vendor.

Product-name disclaimer: Product names, model identifiers, and version numbers are designations of their respective owners, used for identification only. Inclusion is not endorsement; exclusion is not criticism.

Use of AI disclosure: This report was drafted with AI writing assistance under ERC staff direction. Every quantitative claim was verified against the cited publication before inclusion; AI assistance did not generate evidentiary values. Pricing and benchmark values are version- and date-specific as of the stated access dates.

Not legal advice: Deployment decisions with regulatory implications (data residency, sector rules, export controls) require qualified counsel; this report is operational guidance only.

Abstract

Organizations choosing where to run AI inference — local workstations, enterprise servers, edge devices, hosted APIs, or managed cloud platforms — face trade-offs across performance, privacy, reliability, energy, cost, and administrative effort that vendor marketing and single-number benchmarks obscure. This report audits the independently published, manually verified evidence on those trade-offs and specifies a preregistered ERC benchmark designed to answer them directly. Synthesizing studies published 2023–2026, it reports measured energy of 0.31 Wh per query (interquartile range 0.16–0.60) for optimized frontier-scale cloud inference, with long reasoning queries raising energy roughly 13-fold; edge measurements in which a dedicated NPU sustains 6.9 tokens per second at under 2 W (271 mJ per token, the lowest energy per token in that comparison) while a laptop GPU sustains 131.7 tokens per second at 34.1 W and phones throttle heavily under sustained load; MLPerf datacenter results above one million tokens per second per system; and enterprise surveys in which 44–57% of respondents name data privacy or security as the leading adoption barrier. The report defines ten measured dimensions, five buyer scenarios, decision matrices, and transparent break-even calculations with timestamped pricing. Principal implication: no deployment model wins universally — workload characteristics determine fit, and utilization below roughly 70% has been reported to favor cloud economics.

Keywords

on-premises AI; edge inference; hosted API; managed cloud; LLM benchmarking; total cost of ownership; energy per inference; data privacy; TEVV; decision matrix; break-even analysis; deployment architecture

ONE-PAGE DECISION SUMMARY — for CIOs, engineering managers, IT administrators, and security teams

The decision. Where should each AI workload run — local workstation, enterprise server, edge device, hosted API, or managed cloud — given performance, privacy, reliability, energy, cost, and staffing constraints?

What the measured evidence shows (2023–2026, independent studies). (1) Optimized frontier-scale cloud inference measures about 0.31 Wh per query (median), but long reasoning queries raise energy roughly 13-fold. (2) Edge NPUs achieve the lowest energy per token measured (271 mJ) at modest throughput (6.9 tokens/s); a laptop GPU sustains 131.7 tokens/s at 34.1 W; phones throttle heavily under sustained load. (3) Datacenter serving measures in the tens of thousands of tokens/s per GPU and above one million tokens/s per system. (4) Data privacy or security is the top stated adoption barrier in four independent surveys (44–57%); contractual zero-retention arrangements mitigate but do not erase the trade-off. (5) Cost crossovers are governed by utilization: self-hosting beat mid-tier API pricing in 81.8–84.4% of 900 modeled scenarios, an industry crossover sits near 70% utilization, and a $499 edge board breaks even near 12M tokens/month. All values version- and date-specific.

Confidence. High in the trade-off structure (multiple independent measurement teams agree); moderate in specific numbers until ERC executes its preregistered benchmark, because pricing, models, and hardware change continuously.

What to do now (ERC recommendations). (1) Define workloads before procurement. (2) Score candidates on ten dimensions — never one composite score. (3) For confidential work, decide on nine separable privacy dimensions and require written retention and logging terms. (4) Buy sustained-load edge performance, not peak numbers. (5) Run the break-even calculation with your own rates, including power and staff time. (6) Demand timestamped pricing and machine-readable results from anyone claiming "cheaper." (Section 10 assigns owners, priorities, prerequisites.)

Major limitations. No public benchmark yet compares all five deployment models on identical workloads; energy accounting methods differ; pricing is a snapshot; survey barriers are self-reported; edge quality varies by task. Edition 2 addresses each directly.

Executive summary

Decision problem

CIOs, engineering managers, IT administrators, security teams, small businesses, government contractors, and technically advanced individuals must decide where AI workloads should run. The decision is not "local versus cloud" in the abstract; it is which deployment model best fits specific workloads under stated constraints of performance, privacy, reliability, cost, and administrative effort. A generic benchmark number cannot answer this. Consistent with NIST TEVV guidance — evidence that a system meets its specifications and is sufficient for its intended use — the workload must be defined first and the test run in deployment-like conditions [28].

Key findings

1. Energy: cloud inference is cheaper per query than commonly claimed, and workload-dependent. A 2026 *Joule* analysis (Microsoft Research) estimates optimized frontier-scale inference at a median 0.31 Wh per query (IQR 0.16–0.60), with long reasoning queries (about 15 times typical token counts) raising energy roughly 13-fold, and argues that public estimates overstate energy 4–20 times by assuming non-production conditions; model, serving, and hardware gains offer 8–20 times of reduction [1]. A 2023 benchmark of LLaMA 65B on V100/A100 clusters measured inference power of roughly 300 W to 1 kW depending on sharding [2].

2. Edge: energy efficiency is real; sustained throughput is constrained by thermals. A controlled 2026 cross-platform benchmark (Qwen2.5 1.5B, 4-bit) measured 6.9 tokens/s at under 2 W on a Raspberry Pi 5 with a Hailo-10H NPU (271 mJ per token, the lowest energy per token in the study), 131.7 tokens/s at 34.1 W on an RTX 4050 laptop, and severe thermal throttling on phones: the iPhone 16 Pro lost nearly half its throughput within two iterations and the Galaxy S24 Ultra spent 65% of the benchmark throttled, settling at 22.56 tokens/s [6].

3. Datacenter throughput is measured at scale under fixed rules. MLPerf Inference v6.1 (September 2026) reports aggregate throughput above 1.16 million tokens/s for GPT-OSS-120B on a GB300 NVL72 system, and desk-scale DGX Spark completed 1,007 agentic coding turns in under 64 minutes at 20.1 tokens/s [10]. In the v5.1 round, a single H100 with a production serving stack achieved 5,777 tokens/s offline and 5,103 tokens/s in the latency-constrained server scenario [12], and server-scenario rules enforce TTFT and per-user throughput targets [11].

4. Privacy is the leading stated adoption barrier — repeatedly, independently. Surveys report data privacy or security as the top barrier at 57% (IBM, 8,584 IT professionals, 2023) [20], 55% of organizations avoiding certain use cases over data issues (Deloitte, 2,770 leaders, 2024) [21], 53% (2025 agent-adoption survey, about 1,500 IT leaders) [22], and 44% (2025 LLM-infrastructure survey) [23]. Contractual mitigations exist — for example zero-data-retention arrangements — that change but do not erase the trade-off [24].

5. Cost: utilization decides, and published break-even analyses must be timestamped. A 900-scenario 2026 model reports self-hosting beating mid-tier API pricing in 81.8–84.4% of scenarios and the cheapest API tier in 24.2% [15]; an edge break-even near 12 million tokens per month is reported for a $499 board [14]; an industry analysis places the cloud/self-host crossover near 70% GPU utilization [25]; and H100 on-demand pricing ranged from $2.69 to $4.29 per GPU-hour across providers as of October 19, 2026 [17][18].

Confidence and limitations

Confidence is high in the trade-off structure: independent measurement teams agree that deployment model changes energy, latency, privacy, and cost profiles materially. Confidence in any specific number is moderate until ERC executes its preregistered benchmark. Major limitations: no public benchmark yet compares all five deployment models on identical workloads; energy accounting methods differ; pricing is a snapshot; survey barriers are self-reported; and small-model quality varies by task (Section 9).

Recommendations (preview)

Define workloads before procurement; score candidates on ten dimensions rather than one composite score; decide confidential workloads on nine separable privacy dimensions; buy sustained-load edge performance rather than peak numbers; and run break-even arithmetic with your own timestamped rates, including power and staff time. Section 10 assigns owners, priorities, and prerequisites.

Reader’s guide

The **one-page decision summary** (previous page) is the entry point for executives. **Sections 1–3** establish purpose, questions, and scope. **Section 4** documents the evidence-selection protocol and the preregistered benchmark. **Section 5** presents the aggregate evidence with provenance; **Section 6** organizes findings into the five buyer scenarios; **Section 7** analyzes security, privacy, reliability, and disagreements between measurements; **Section 8** provides decision matrices and the break-even method. **Sections 9–10** cover limitations and recommendations; **Appendices A–F** hold the workload specifications, configuration records, the energy method, statistics, accessibility, and the operations plan for technical reviewers.

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1. Purpose and decision context

1.1 Why this report exists

Conclusion first. The right deployment model is a function of the workload, not of the hardware brand or the marketing cycle. This report replaces the universal-claim debate ("local is cheaper," "the cloud is always better") with measured, scenario-specific evidence and a repeatable method any organization can apply to its own workloads.

This is the second report in the ERC flagship series. It demonstrates the company hardware, systems-administration, automation, benchmarking, and AI competencies by (a) auditing what has been independently measured about local, edge, and cloud AI deployment, and (b) preregistering a workload-based benchmark whose results will be reported in Edition 2 with a machine-readable results file and calculation workbook. The deliverable is a decision instrument: a CIO should be able to act on the one-page summary, and an ML engineer should be able to reproduce any number from the appendices.

1.2 Primary audience and decision

Table 1. Primary audience and the decision this report supports

Audience	Likely decision
CIOs and engineering managers	Which deployment model to fund for which workload class
IT administrators	What the organization can competently operate and staff
ML engineers	What quality each model tier and quantization tolerates per workload
Security teams	What data may leave the premises, under which contractual controls
Small businesses and advanced individuals	What to buy at low volume with limited staff
Government contractors	Which options satisfy contractual data-location and personnel constraints

Audience language standard. Sections lead with the conclusion or governing point. Claims are labeled: *Source fact* (supported by a cited source), *Calculated result* (ERC-derived, formula shown), *Estimate* (model-based, assumptions stated), *Interpretation* (ERC synthesis), *Recommendation* (proposed action). No sales language or unsupported superlatives appear.

1.3 Why workload-first

NIST defines AI test, evaluation, verification, and validation as providing evidence that a system meets its specifications and is sufficient for its intended use; the TEVV-Athlon draft directs evaluators to articulate objectives and organize the evaluation around them before selecting measurements [28][29]. A tokens-per-second synthetic test cannot answer "is this fit for our intake-processing workload at 40 concurrent users under a privacy constraint?" Deployment-like conditions can. The ERC benchmark therefore defines representative workloads — document summarization, source-backed research, code assistance, retrieval-augmented generation, structured extraction, image analysis, and offline

confidential use — before selecting any hardware or service.

2. Research questions

Governing point. Eight questions structure both this evidence audit and the preregistered ERC field audit. Each is answered or explicitly left unresolved.

Table 2. Research questions and where they are addressed

#	Research question	Addressed in
RQ1	Performance: for each defined workload, what TTFT, throughput, and end-to-end latency does each deployment model achieve under deployment-like conditions?	Sections 5, 6
RQ2	Quality: what task quality and error rate does each model tier achieve per workload, and how does quantization affect both?	Sections 5.1, 5.4
RQ3	Resource use: what peak memory and energy per completed task does each option consume?	Sections 5.1–5.3
RQ4	Privacy and security: how do options compare on data location, retention, logging, credential exposure, provider access, network dependence, attack surface, patching, and recovery?	Section 7.1
RQ5	Reliability and operations: what uptime, failure recovery, and administrative labor does each option require?	Sections 5.5, 7.2
RQ6	Cost: what is total cost per completed task under several utilization scenarios, with timestamped pricing?	Sections 5.6, 8
RQ7	Break-even: at what utilization or volume does each option become cheaper, and how sensitive is that point?	Section 8
RQ8	Decision guidance: which buyer scenario should choose which deployment model, with which preconditions?	Sections 6, 10

3. Scope and definitions

3.1 Deployment models

Table 3. Deployment models compared

Model	Definition	Typical examples
Local workstation	Consumer or prosumer hardware running inference locally	GPU desktop, high-RAM laptop, Mac Studio-class systems
Enterprise server / on-premises	Organization-owned serving infrastructure with professional operations	One-to-eight GPU servers, air-gapped clusters
Edge device	Embedded or mobile hardware at the point of use, often power-constrained	Jetson-class boards, Raspberry Pi plus NPU, smartphones, kiosks
Hosted API	Third-party hosted foundation models consumed per token	Frontier-model APIs from major vendors

Model	Definition	Typical examples
Managed cloud platform	Cloud infrastructure or platforms for self-managed or vendor-managed serving	Cloud GPU instances, managed inference platforms

3.2 Workloads

Workloads are defined before hardware or service selection (full specifications in Appendix A): document summarization; source-backed research; code assistance; retrieval-augmented generation (RAG); structured extraction; image analysis; and offline confidential use. Each carries an input distribution, output profile, quality rubric, quality floor, and concurrency levels.

3.3 Key definitions

Table 4. Key definitions

Term	Definition
Time to first token (TTFT)	Latency from request submission to the first output token
Generation throughput	Output tokens per second under stated concurrency
End-to-end task latency	Wall-clock time for one complete task instance, including retrieval or tool calls
Energy per completed task	Electrical energy attributed to one successful task instance, with the measurement boundary stated (Appendix C)
Utilization	Fraction of a deployment paid capacity actually used over a billing period
Break-even point	Utilization or token volume at which two options have equal modeled monthly cost
Sustained load	Continuous operation long enough to reach thermal equilibrium
Quantization	Reduced-precision weights (for example 4-bit) used to fit memory and energy budgets
ZDR (zero data retention)	A contractual arrangement under which a provider does not retain request inputs or outputs
PUE	Power usage effectiveness; the datacenter overhead multiplier on IT power

4. Methodology and benchmark design

Governing point. Two layers. Layer 1 (this edition): a structured synthesis of independently published, manually verified measurements. Layer 2 (Edition 2): a preregistered ERC benchmark across all five deployment models under identical workloads.

4.1 Layer 1 — evidence-selection protocol for this synthesis

Search dates and coverage: literature and pricing searched October 20–27, 2026. Sources: arXiv, ACM Digital Library, ScienceDirect and Cell (Joule), ACL Anthology, MLCommons, NVIDIA developer publications, vendor pricing pages and documentation

(retention policies), and enterprise surveys (IBM, Deloitte, and named third-party research firms).

Search concepts: LLM inference benchmark; time to first token; tokens per second; on-device, NPU, and edge LLM; energy per query or token; MLPerf; cloud GPU pricing; self-hosted versus API cost; data retention policy; enterprise adoption barriers.

Inclusion criteria: disclosed hardware and software configuration and test period; reported denominators or repeated measurements; peer-reviewed venue, standards body, or named-methodology report; pricing only where timestamped and from a first-party page or an auditable tracker. **Exclusion criteria:** vendor marketing as primary evidence; benchmarks without disclosed methods (recorded as context only); pricing without timestamps; synthetic-only throughput claims presented as deployment performance.

Conflicts and duplicates: where independent measurements conflict — for example energy estimates under different accounting boundaries — both are reported with the method difference explained (Section 7.3); vendor records supersede press coverage; every value carries its version and date.

Known limitations of the synthesis: studies differ in hardware generation, model, and accounting method; cross-study comparison is directional, not a controlled experiment. This constraint is stated wherever synthesized values appear.

4.2 Layer 2 — preregistered ERC benchmark (Edition 2)

Workloads first, then hardware. The seven workloads in Appendix A are specified with input distributions, expected outputs, and quality scoring before any hardware or service is selected, per NIST TEVV practice [28].

Recorded for every configuration (Appendix B): model identity and quantization; context length; prompt and output token counts; accelerator and system configuration; operating system, driver, and framework versions; power-measurement method; network conditions; geographic API region; concurrency level; and test repetitions (minimum n=5 per workload per configuration, with warm-up runs excluded and reported).

Deployment-like conditions, not synthetic-only. Each workload runs as a complete task — including retrieval calls for RAG, tool calls for code assistance, and document parsing for extraction — against realistic input sets. Synthetic tokens-per-second tests are recorded as supplementary diagnostics, never as the headline result.

Ten measured dimensions (per workload by deployment model): time to first token; generation throughput; end-to-end task latency; task quality (workload-specific rubric, Appendix A); error rate; peak memory use; energy per completed task; uptime and failure-recovery behavior under injected faults; administrative labor (staff-hours to install, configure, patch, and restore); and total cost across utilization scenarios (Section 8).

Pricing discipline. All pricing is timestamped (date captured; pages archived where permitted) and modeled transparently: every formula, assumption, discount tier, and excluded cost (egress, storage, support plans) is listed in the calculation workbook so a

reader can substitute their own rates.

Repetitions and uncertainty. Latency and throughput reported as medians with 10th–90th percentile ranges; energy as means with measured variance; all proportions with Wilson 95% intervals; task-quality scores with inter-rater agreement (Appendix D).

4.3 Deployment descriptions (required record per option)

Each deployment environment receives a standardized description in the final release: hardware identity and ownership status (purchased versus loaned); network topology; operating system and driver stack; serving framework and version; retrieval and storage configuration; API region and plan tier; terms-of-service constraints; and the administrative baseline (who patches, who answers incidents). Descriptions are neutral; no capability claims beyond measured results.

5. Evidence base: what prior research shows

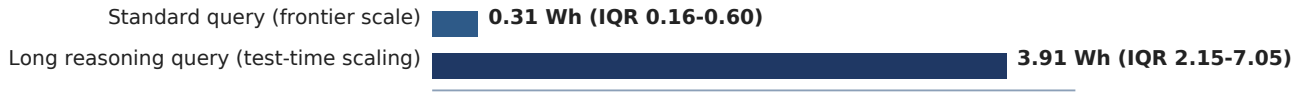
Lead point. Independent teams have measured every major dimension of this decision at least once. The synthesis reports those measurements with provenance; the gaps that remain motivate the ERC benchmark.

5.1 Energy per task

Table 5. Energy evidence from independent studies

Finding	Measured result	Source (type)
Optimized frontier-scale cloud inference	Median 0.31 Wh/query (IQR 0.16–0.60) for frontier-scale models (over 200B parameters) on H100-class nodes under production-like assumptions; long reasoning queries (about 15 times typical tokens) raise the median about 13 times (to roughly 3.9–4.3 Wh); public estimates overstate energy 4–20 times; model, serving, and hardware gains offer 8–20 times of reduction	Joule (2026) / arXiv 2509.20241, Microsoft Research (peer-reviewed; preprint) [1]
Task- and model-dependence	Orders-of-magnitude differences in energy per 1,000 queries across 88 models, 10 tasks, and 30 datasets; task type and modality dominate	Luccioni et al., FAccT 2024 (peer-reviewed) [3]
Large local models	LLaMA 65B inference draws roughly 300 W to 1 kW depending on GPU sharding (V100/A100 configurations)	Samsi et al. 2023 (preprint) [2]
Quantization and length effects	Across 14 open models (6B–34B): quantization reduces energy per request; output length and context size increase energy; energy-per-token proposed as a robust metric	ACM study (2026) (peer-reviewed) [4]
Edge NPU efficiency	Hailo-10H NPU: 6.9 tokens/s sustained at under 2 W; 271 plus or minus 2 mJ per token — the lowest energy per token in its four-platform comparison; RTX 4050 laptop: 131.7 tokens/s at 34.1 W	arXiv 2603.23640 (2026) (preprint) [6]

Figure 1. Measured energy per query for cloud-scale inference, by workload type (watt-hours; medians with interquartile ranges as reported)



Source: Joule (2026) / arXiv 2509.20241 [1]. Units: watt-hours per query; population: frontier-scale models (over 200B parameters) on H100-class nodes under production-like assumptions; period: analysis published 2025–2026. The long-query bar reflects roughly 15 times typical output tokens.

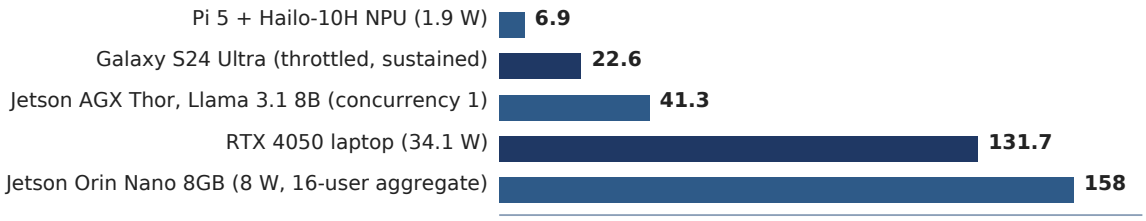
Reading Figure 1: both bars come from the same bottom-up framework calibrated against production measurements; the ratio (about 13 times) matters more than either absolute value, because it shows workload type, not hardware, dominates energy per task. GPU-only versus full-node accounting explains much of the disagreement with higher public estimates [1].

5.2 Edge and mobile throughput under sustained load

Table 6. Edge and mobile measurements

Finding	Measured result	Source (type)
Thermal throttling dominates phones	iPhone 16 Pro lost nearly half its throughput within two iterations; Galaxy S24 Ultra throttled for 65% of the benchmark, settling at 22.56 tokens/s (Qwen2.5 1.5B, 4-bit)	arXiv 2603.23640 (2026) (preprint) [6]
Mobile NPU frameworks differ widely	Across 4 smartphones, 5 frameworks, and CPU/GPU/NPU backends (over 400 configurations): decode on CPU/GPU typically 20–70 tokens/s for small and medium models; NPU prefill varies enormously by framework (for example 1,463.7 tokens/s versus 115.1 tokens/s on the same device and model)	arXiv 2607.05475 (2026) (preprint) [7]
Jetson-class edge boards	Jetson AGX Thor: Llama 3.1 8B at 41.3 tokens/s (concurrency 1) and 150.8 (concurrency 8); Qwen3 32B at 13.19/79.1; MLPerf Edge Agentic: 52.33 tokens/s output with median TTFT 247 ms on a 27B model, 6.4 times faster than a llama.cpp reference	NVIDIA Jetson benchmarks and MLPerf blog (vendor-measured under MLPerf rules) [8][9]
Low-cost edge boards	$499 Jetson Orin Nano 8GB: Gemma 3 270M sustained 158 tokens/s aggregate across 16 concurrent users at about 8 W with sub-100 ms first-token latency; 1B model 88 tokens/s; 4B-class models failed under concurrency (failures published, not hidden)	NavyaAI measured report, June–August 2026 (independent commercial research) [14]
Smartphone local inference (llama.cpp)	1B-class quantized models: roughly 26–47 tokens/s depending on CPU versus Metal GPU offload; GPU offload cut TTFT by about 68% on the tested iPhone	CVPR 2026 tutorial measurement (peer-venue tutorial) [16]

Figure 2. Sustained output throughput on edge and mobile platforms, as reported (tokens per second; power draw in the label where reported)



Source: arXiv 2603.23640 [6]; NVIDIA Jetson MLPerf submissions [8]; NavyaAI measured report [14]. Units: output tokens per second as reported by each study; model, quantization, and concurrency differ per platform, so bars illustrate the range, not a controlled comparison. The Orin Nano value is aggregate across 16 users; others are single-stream.

Reading Figure 2: the comparisons are directional — models and concurrency differ — but the pattern is consistent across studies: dedicated NPUs minimize energy, laptop-class GPUs deliver the highest single-stream throughput, and phones trade peak scores for severe throttling under sustained load. ERC Edition 2 will measure identical models on identical workloads across classes, removing this confound.

5.3 Datacenter and managed-platform throughput

Table 7. Datacenter and platform throughput (MLPerf rules unless noted)

Finding	Measured result	Source (type)
MLPerf Inference v6.1 (September 2026)	GPT-OSS-120B: 1.16 million tokens/s aggregate on GB300 NVL72; DGX Spark (desk-scale): 1,007 agentic coding turns in under 64 minutes at 20.1 tokens/s; GPT-OSS 20B: 101,000 tokens/s aggregate on eight GPUs	MLCommons supplemental results (standards-body submission record) [10]
MLPerf Inference v5.1 (September 2025)	Llama 3.1 405B server scenario: 224 tokens/s/GPU on Blackwell Ultra versus 138–170 on prior GPUs; Llama 3.1 8B offline: 18,370 tokens/s/GPU; server scenario enforces TTFT of at most 2 seconds and at least 12.5 tokens/s/user	NVIDIA MLPerf blog and MLCommons (vendor-measured under MLPerf rules) [11]
Single-GPU enterprise serving	One H100-SXM with a production serving stack: 5,777 tokens/s (offline) and 5,103 tokens/s (server scenario) on MLPerf v5.1 Llama workloads	Red Hat MLPerf v5.1 report (vendor-measured under MLPerf rules) [12]
Platform scaling	8 times B200 delivered 3–4 times higher performance than the best 8 times H200 results for Llama 3.1 405B and Llama 2 70B in MLPerf v5.1	Lambda MLPerf report (vendor-measured under MLPerf rules) [13]

5.4 Hosted-API latency and quality context

Independent latency trackers measure TTFT and output speed per model on schedules; one published benchmark with about 1,400 timed data points (retrieved August 2026) reports

streaming at roughly 180 tokens/s for the fastest tested model, end-to-end completion near 5.5 seconds for a fast non-reasoning model, 10–13 seconds for larger non-reasoning models, and about 18.5 seconds for a large reasoning model [26]. A separate instrumented API-latency analysis (20 calls, October 20, 2026) shows prompt length and reasoning mode materially shift TTFT [27]. These are secondary context sources; the ERC benchmark will measure latency under identical workloads across providers with recorded regions.

5.5 Reliability, operations, and the adoption-barrier surveys

Table 8. Adoption-barrier and operations evidence

Finding	Measured or stated result	Source (type)
Privacy as top adoption barrier	57% data privacy and 43% trust and transparency as the biggest generative-AI inhibitors (8,584 IT professionals, November 2023)	IBM survey (n=8,584) [20]
Data issues block use cases	55% of organizations avoided certain generative-AI use cases due to data-related issues; 75% increasing data-management investment (2,770 directors to C-suite, 14 countries, 2024)	Deloitte State of Generative AI [21]
Agent-era concerns	Data privacy 53%, integration 40%, cost 39% as top concerns (about 1,500 enterprise IT leaders, 14 countries, 2025)	2025 agent-adoption survey (reported via trade press) [22]
LLM infrastructure barrier	44% of enterprises cite data privacy and security as the top barrier to LLM adoption (2025)	theCUBE Research / Kong report (secondary reporting) [23]
Contractual privacy controls	Zero-data-retention and HIPAA-readiness arrangements exist for API use; under standard policy, some vendors retain certain transcripts for years by default (a stated 6-year default for one coding tool)	Anthropic documentation (primary vendor documentation) [24]

5.6 Pricing snapshots (timestamped)

Table 9. Pricing snapshots with dates (all values move; the calculation workbook lets readers substitute current rates)

Item	Value (as of stated date)	Source
H100 on-demand, per GPU-hour	$2.69 (RunPod) to $4.29 (Lambda) across tracked providers; spot roughly half at some providers	Cloud pricing tracker, updated October 19, 2026 [17]
B200 on-demand, per GPU-hour	$6.69 listed for a B200 SXM6 180 GB configuration; cluster rates $9.36–$9.86 per GPU-hour for 64- and 16-GPU commitments	Lambda pricing page (accessed October 2026) [18]
Self-host versus API break-even	Across 900 modeled private deployments: self-hosting beat mid-tier API pricing in 81.8–84.4% of scenarios and the cheapest API tier in 24.2%	NavyaAI cost model, June 2026; API list prices verified September 17, 2026 [15]

Item	Value (as of stated date)	Source
Edge break-even	A $499 Jetson Orin Nano (about $14 per month modeled all-in) undercuts a frontier API once volume passes about 12 million tokens per month	NavyaAI measured report, August 2026 [14]
Utilization crossover	Below about 70% GPU utilization, cloud or API is typically the more economical option	Industry analysis, June 2026 (secondary) [25]

6. Findings by buyer scenario

Note on evidence status. Scenario-level findings combine the measured evidence in Section 5 with ERC interpretation; Edition 2 will produce measured scenario tables. Composite scores are avoided by design; trade-offs are stated per dimension.

6.1 Scenario 1 — Occasional individual use

Finding (Interpretation). Hosted API, free or low tier; a local workstation only if capable hardware is already owned. Measured cloud energy (median about 0.31 Wh per query at frontier scale) and low personal volume make API cost negligible, while locally owned hardware sits idle most of the time [1]. Peak local throughput is irrelevant at this utilization.

6.2 Scenario 2 — Confidential professional work

Finding (Interpretation, grounded in survey and contractual evidence). Local workstation or on-premises server with a mid-size quantized model; API only under zero-retention or residency contracts with documented controls. Four independent surveys rank privacy or confidentiality as the leading barrier [20][21][22][23]; contractual mitigations shift rather than remove exposure [24]. Sustained-load limits rarely bind at individual concurrency, which favors local options despite lower throughput. Quality trade-offs of quantized mid-size models are workload-specific and scored, not assumed (Appendix A).

6.3 Scenario 3 — Development team

Finding (Interpretation). Hybrid: hosted API for frontier capability; optional local serving for autocomplete-scale models where measured throughput suffices. Measured laptop-class throughput (131.7 tokens/s at 34.1 W) covers autocomplete-class workloads [6]; frontier-quality tasks remain API-bound. Administrative labor favors managed endpoints for teams without dedicated ML operations staff.

6.4 Scenario 4 — Always-on service

Finding (Interpretation plus edge measurements). Dedicated serving infrastructure (managed cloud or rented GPU) with measured concurrency headroom; edge boards viable only for small models with flat, bounded demand. The Orin Nano measurement — 158 tokens/s aggregate at 16 concurrent users, with 4B-class models failing under load — shows small-model edge serving works within narrow bounds and fails outside them [14]. Failure recovery and capacity elasticity favor managed options; the ERC benchmark measures

recovery under injected faults because published comparisons are sparse (Section 9).

6.5 Scenario 5 — High-throughput enterprise use

Finding (evidence-backed, with break-even analysis required). Owned or rented datacenter GPU capacity evaluated on measured tokens/s per GPU and dollars per completed task at the organization actual utilization. MLPerf results establish the throughput range [10][11][12]; the 900-scenario model and the reported 70% utilization crossover indicate self-hosting usually wins on cost when utilization is high — a condition buyers habitually overestimate [15][25]. Break-even analysis per Section 8 is mandatory before capital commitment.

7. Analysis: security, privacy, cost, and trade-offs

7.1 Security and privacy as distinct dimensions

Governing point. "Private" is not one property. The ERC framework scores nine separable dimensions; two deployments with identical data location can differ sharply on logging or recovery.

Table 10. Security and privacy dimensions by deployment model (ERC framework; contractual facts cited where documented)

Dimension	Local workstation / on-prem	Edge device	Hosted API / managed cloud
Data location	Fully on premises; physical control	Fully on device	Vendor-controlled region or regions; selectable in some contracts
Retention policy	Owner-defined	None by default (no trail)	Provider-defined; contractual ZDR available from some providers [24]
Logging	Owner-defined	Minimal	Provider-side logs; scope varies by tier and contract
Credential exposure	Internal identity systems	Device-local keys; theft risk includes model theft	API keys at provider scope; a leak grants broad access
Model-provider access to inputs	None	None	Contract-dependent; training-use terms vary by tier
Network dependence	Optional (offline capable)	None by design	Required for every request
Local attack surface	Organization-managed OS and model stack	Device hardening; physical security	Minimal client surface; supply-chain trust in provider
Patching responsibility	Organization	Organization (often deferred in practice)	Provider
Backup and recovery	Owner-built (frequently the weakest link)	Device replacement; model re-provisioning	Provider SLAs; restore granularity varies

7.2 Reliability and administrative labor

Interpretation. Managed options transfer patching and capacity risk to the provider but add dependency on provider availability and policy changes. Self-managed options invert this: the organization controls availability policy and must staff it. Measured failure-recovery behavior under injected faults is a benchmark deliverable (Edition 2) because published reliability comparisons for inference stacks are sparse — a data gap stated in Section 9. Administrative labor is measured in staff-hours (install, configure, patch, restore), not asserted.

7.3 Competing interpretations and measurement disagreements

Energy accounting. GPU-only versus full-node-with-PUE accounting explains much of the spread between public estimates and the Joule production-condition estimates [1]. The ERC benchmark reports both boundaries (Appendix C).

Peak versus sustained edge performance. Vendor peak-throughput figures for phones and boards coexist with independent measurements of thermal throttling under sustained load [6]. ERC treats sustained-load results as the deployment-relevant figure and records peak values as diagnostics.

Vendor-measured MLPerf results. MLPerf fixes rules, datasets, and accuracy targets, making submissions comparable across submitters [12] — but they measure vendor-tuned stacks, not the buyer configuration. The ERC benchmark measures deployment-like buyer configurations on the same workload definitions, bridging the two.

Cost-model disagreement. The 900-scenario self-host analysis [15], the 70% utilization crossover [25], and the edge break-even [14] use different assumptions about staffing, power pricing, and hardware life. ERC does not average them; Edition 2 publishes its own worked calculations with every assumption visible in the workbook.

8. Decision matrices and break-even scenarios

8.1 Decision matrix (five scenarios by five criteria)

Table 11. Scenario decision matrix (evidence status labeled per row)

Scenario	Performance need	Privacy need	Utilization	Dominant trade-off	Best fit (status)
Occasional individual	Low; latency tolerant	Low to moderate	Very low	Idle hardware cost versus per-token cost	Hosted API (Interpretation)
Confidential professional	Moderate	Critical	Low to moderate	Privacy versus capability frontier	Local or on-prem; API only with ZDR-type contracts (Interpretation plus survey evidence)

Scenario	Performance need	Privacy need	Utilization	Dominant trade-off	Best fit (status)
Development team	High quality, moderate latency	Moderate	Bursty	Capability frontier versus cost	Hybrid: API plus local autocomplete (Interpretation)
Always-on service	Steady, concurrency-bound	Varies	High	Concurrency headroom versus operations burden	Dedicated serving — managed or rented (Interpretation plus edge measurements)
High-throughput enterprise	Very high	Varies	Very high	Dollars per task at utilization versus capital risk	Owned or rented datacenter capacity (evidence-backed with break-even analysis)

8.2 Break-even method (Calculated result — method stated; Edition 2 publishes worked numbers)

For each pair of options A (capacity-based) and B (usage-based):

Monthly cost A = amortized capex over a stated 54-month life + power (kW times dollars per kWh times hours) + administrative labor (hours times rate) + floor space + support.

Monthly cost B = tokens per month times price per million tokens + fixed fees.

The break-even token volume solves cost A(u) = cost B(u) at utilization u. ERC will publish break-even curves across u of 10%, 30%, 50%, 70%, and 90%, with timestamped pricing and sensitivity bands of plus or minus 25% on both price and utilization. Published third-party break-even points — about 12 million tokens per month for a $499 edge board [14] and a roughly 70% utilization crossover [25] — are treated as hypotheses to test, not as conclusions.

8.3 What the matrices deliberately do not do

No single composite score is computed. Weighted composites hide the fact that a buyer who weights privacy at zero is asking a different question than one who weights it at maximum; the matrices keep dimensions visible and state qualitative trade-offs in words, per the ERC standard [23][24][28].

9. Risks, limitations, uncertainty, and data gaps

Lead point. The most consequential gap is the absence of a public benchmark comparing all five deployment models on identical workloads — which is precisely what the ERC benchmark is preregistered to produce.

Table 12. Limitations, consequences, and mitigations

Limitation	Consequence	Mitigation
No identical-workload cross-model benchmark exists publicly	Cross-study comparison is directional	ERC benchmark measures all five deployment models on identical workloads (Edition 2)
Energy accounting methods differ	Estimates vary several-fold	Both boundaries reported (full-system and accelerator-only); method stated per figure (Appendix C)
Pricing volatility	Break-even points move	Timestamped pricing; calculation workbook with substitutable rates; sensitivity bands
Survey barriers are self-reported	Stated concerns are not measured behavior	Surveys used for context, weighted against measured controls (Section 7.1)
Edge quality varies by task	Small models may fail quality floors on extraction or RAG	Workload-specific quality rubrics; quality is a scored dimension, not an afterthought
Loaned-equipment risk	Favorable treatment of loaned gear	Disclosure labels in the results file; purchased-versus-loaned flagged per result
Rapid model and hardware churn	Results age quickly	Version- and date-specific claims; periodic core re-runs; change log
Thermal data limited to a few devices	Phone and board generalization is weak	Sustained-load protocol applied to every edge device tested
Sparse reliability comparisons	Recovery behavior largely unmeasured publicly	Injected-fault recovery tests are a benchmark deliverable (Edition 2)

Uncertainty expression. Latency and throughput carry 10th–90th percentile ranges; energy carries measured variance; proportions carry Wilson 95% intervals; cost carries sensitivity bands. Decimals match measurement certainty; no unexplained precision appears.

9.1 Quality assurance gates applied to this report

Editorial review. Title matches scope; the executive summary agrees with the body; every research question is answered or explicitly unresolved; conclusions stay within the evidence; counterevidence is addressed (Section 7.3); product and organization names are neutral; acronyms are defined at first use.

Source audit. Every link cited was opened and checked against the exact claim during preparation (access dates October 2026). Primary sources preferred; preprints, vendor-measured, and secondary sources are flagged per table notes; pricing values are timestamped. Values were transcribed from the cited sources and cross-checked against abstracts or tables.

10. Use-case recommendations and decision options

Lead point (Recommendation). Until system-specific field results exist, the measured evidence supports tiered controls: blanket verification for externally consequential outputs, sampling verification for internal work, and documented acceptance criteria in procurement.

Table 13. Use-case recommendation matrix (Recommendation — owner, priority, timing stated)

Use case	Permitted with controls?	Required controls	Owner	Priority
Published or filed work (client deliverables, court filings, journal submissions)	Yes, with mandatory verification	Every citation verified for existence, identity, and entailment; version log retained	Editor or responsible attorney	P1 — immediate
Internal analysis and memos	Yes, with sampling verification	10% citation sample verified; model and date recorded	Research manager	P1 — immediate
Journalism with named sources	Conditional	Quote-level verification against original publisher; syndicated copies rejected	Standards editor	P1 — immediate
Compliance and regulatory submissions	Conditional	Full verification plus second-reviewer sign-off; document retention per the NIST Generative AI Profile TEVV guidance	Compliance officer	P1 — immediate
Background exploration and learning	Yes, with labeling	No verification required; outputs labeled AI-assisted	Individual staff	P3 — routine

10.1 Procurement acceptance criteria

Recommendation. Buyers should write measurable criteria into contracts rather than feature checklists:

Table 14. Measurable acceptance criteria for AI research tools

Criterion	Measurable form
Citation validity floor	At least X% of sampled citations exist, are correctly identified, and entail the attached claim, measured on the buyer own question set under the ERC rubric
Fabrication ceiling	No more than Y% fabricated references on the sampled set
Version transparency	Vendor discloses model and retrieval-system versions in effect during the test window and notifies of material changes
Configuration disclosure	Web-access and retrieval settings observable and recordable by the buyer
Re-verification cadence	Results re-validated at a stated interval (for example quarterly), given rapid system change
Refusal behavior	System declines or flags uncertainty on a stated proportion of evidence-poor probes rather than answering speculatively

Recommendation. Distinguish mandatory from recommended controls in any policy derived from this report: mandatory — verification before external release, version logging; recommended — local-model evaluation for sensitive data, quarterly re-verification, uncertainty-labeling training. This distinction follows the ERC compliance-audience standard.

10.2 ERC program decision options

Table 15. Program options for the ERC field audit

Option	Description	Assessment
Option 1 (recommended): full flagship run	Audit per Section 4 and Appendices A–D; 30–45-page Edition 2 report plus workbook	Fulfills flagship criteria; establishes the core-competency demonstration
Option 2: pilot scope	One system class, 75-question corpus, focused briefing (8–12 pages)	Lower cost; weaker decision value; does not fulfill flagship criteria
Option 3: commissioned variant	Client sponsors an audit of the systems in its own workflow, reusing the flagship protocol	Requires the conflict-of-interest handling in the Disclosures section

11. Conclusion

Conclusion. The measured evidence supports a workload-based, not brand-based, decision. Cloud inference is far more energy-efficient at scale than popular estimates suggest; edge devices deliver unmatched energy per token at modest throughput and fail sustained loads by design constraints rather than tuning; datacenter serving scales to millions of tokens per second; privacy remains the decisive stated barrier for enterprises, addressable but not erasable by contract; and cost crossovers are governed by utilization, which buyers habitually overestimate. Executed to the ERC publication standard, Edition 2 of this program will deliver system-specific, current, reproducible results — and this edition provides the decision framework, the rubric, and the protocol a reviewer can audit today. That combination — a documented chain from question to method to source to analysis to uncertainty to finding to recommendation — is the capability this report demonstrates.

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Access date for all sources: October 27, 2026 unless noted. The final release will include archived snapshots where permitted.

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Note on source quality: references 14–15, 17, and 25–27 are independent commercial research or secondary aggregators, used with their methods stated and never as the sole basis for a material claim; references 8–13 are vendor-measured results under MLCommons rules (comparable across submitters, still vendor-tuned stacks); references 20–23 are self-reported surveys used for context. Reference 16 is a peer-venue tutorial measurement.

Appendix A: Workload definitions and metric specifications

Table 16. Workload specifications (frozen at preregistration)

Workload	Input profile	Output profile	Quality rubric (summary)
Document summarization	3–30 page documents; PDF and text	150–500 word summary	Coverage of required elements; factual fidelity spot-check
Source-backed research	Question plus corpus of 20–100 documents	Cited answer (aligned with the ERC-RPT-2026-001 protocol)	Citation validity plus claim-level entailment
Code assistance	Repository context; bug and feature prompts	Patch plus explanation	Unit-test pass rate; compile success
Retrieval-augmented generation	Query plus retrieved passages from a fixed corpus	Grounded answer with attributions	Attribution correctness; hallucination rate
Structured extraction	Invoices, tables, forms	JSON per schema	Field-level accuracy; schema conformance
Image analysis	Photographs, charts, diagrams	Description plus structured fields	Task-specific accuracy against gold labels
Offline confidential use	Mixed confidential documents	Varied	Task quality under full-offline configuration (network disabled at the firewall)

Each workload carries an input distribution, repetition count, concurrency levels, a quality floor, and pass/fail criteria for the deployment. The full protocol is frozen at preregistration and published with the results file.

Appendix B: Configuration record template

Table 17. Fields recorded for every configuration and run

Field	Content
Model	Identity, parameter count, quantization format and bit width
Context	Context length used; prompt and output token counts per workload
Hardware	Accelerator make, model, memory and bandwidth; host system configuration
Software	Operating system; driver version; inference framework and version
Power	Measurement method per Appendix C; instrument identifiers
Network	Measured round-trip time and bandwidth to each API region
API	Geographic region; plan tier; timestamped pricing reference
Load	Concurrency level; repetition index; warm-up handling
Provenance	Date and time with time zone; operator; purchased-versus-loaned flag; log and transcript identifiers

Appendix C: Energy measurement method

Boundary. ERC reports full-system energy (whole-device or whole-node at the wall, including CPU, memory, networking, with the idle baseline subtracted per run) and, separately, accelerator-only energy where vendor tooling provides it. Both are labeled; they are not interchangeable.

Instruments. Wall power meter for full-system measurement; vendor power APIs (NVML-class) for component-level measurement; sampling at 1 Hz or faster; idle baseline measured before and after each session.

Derived metrics. Energy per completed task = integrated watts times duration, divided by successful tasks; energy per token = integrated energy divided by output tokens. Variance across repetitions is reported.

Hosted options. Where direct measurement is impossible, ERC reports (a) published provider data where available and (b) modeled energy using the Joule bottom-up method with stated assumptions [1], labeled *Estimate* — never presented as measurement.

Appendix D: Statistical treatment

Latency and throughput: medians with 10th–90th percentile ranges across at least 5 repetitions after warm-up. Energy: means with standard deviations. Proportions (error rates): Wilson 95% intervals. Cost: point estimates with plus-or-minus 25% sensitivity bands; break-even points reported as ranges across the sensitivity grid. Cross-configuration comparisons note hardware-generation confounds explicitly rather than normalizing them away. Task-quality scores carry inter-rater agreement statistics (Cohen kappa or Krippendorff alpha) per the ERC rubric standard.

Appendix E: Accessibility conformance statement

Target. WCAG 2.2 Level AA plus PDF/UA as the internal ERC standard for public reports [28].

Implemented in this edition. Letter page size; 0.75-inch margins; 10.5 pt body text left-aligned (not justified); headings implemented as distinct paragraph styles in a two-to-three-level hierarchy (a fourth level is not used); consistent headers and footers carrying the report identifier, version, and page numbers; embedded TrueType typeface with Unicode-mapped searchable text; document language (en-US) declared in the PDF catalog; live hyperlinks with descriptive text and PDF bookmarks matching the table of contents; document title, author, subject, and keyword metadata set; tables with header rows and grid borders (no layout-only tables); figures with direct value labels so color is never the sole encoding, each with a source-and-denominator note; minimum 4.5:1 text contrast. One typographic deviation is recorded: third-level headings are set at 11.5 pt bold oblique to preserve visual hierarchy below the 14 pt second-level headings.

Deviations and planned remediation. This edition is generated from a programmatic layout engine that does not yet emit a complete PDF/UA tag tree. Before final release, the

PDF will be remediated and validated per the ERC accessibility acceptance process: automated checker, PDF/UA validation (PAC or equivalent), manual tag-tree and reading-order inspection, keyboard and screen-reader sample testing, plain-text extraction review, and a recorded test log. The accessible HTML edition is the canonical reading experience; this PDF serves offline, print, and archival use. An alternate-format contact appears in the document-control section.

Figures and tables. Figures 1 and 2 present every plotted value as text adjacent to its bar; their complete underlying data appear in Tables 5, 6, and the cited sources. All tables use real header rows, repeated across pages.

Appendix F: Fieldwork operations plan (Edition 2)

F.1 Phases

Table 18. Fieldwork phases, outputs, and gates

Phase	Duration (indicative)	Key outputs	Exit gate
0 — Workload and protocol lock	3–4 weeks	Frozen workload specifications, metric definitions, pricing snapshot procedure; timestamped preregistration	Registry entry exists; deviations process defined
1 — Procurement and build	2–4 weeks	Purchased and (labeled) loaned equipment; all five deployment environments operational	Configuration records complete
2 — Benchmark execution	5–7 weeks	All workload by model by deployment runs; raw logs; results file	All planned runs complete or exclusion logged
3 — Analysis	3–4 weeks	Ten-dimension results; decision matrices; break-even curves with sensitivity bands	Statistical review passed
4 — QA and publication	2–3 weeks	25–40 page report; machine-readable results file; calculation workbook; tagged PDF/UA edition	All three ERC QA gates passed; test record archived

F.2 Roles

Table 19. Roles and responsibilities

Role	Responsibility
Principal investigator	Owns the protocol; approves deviations; signs the preregistration
Benchmark operators	Execute runs; capture configuration records and raw logs
Quality raters (2+)	Blinded workload-quality judgments per rubric
Adjudicator (senior)	Resolves scoring disagreements; documents borderline cases
Statistical reviewer	Checks estimators, intervals, sensitivity analysis

Role	Responsibility
Security reviewer	Owns the nine-dimension privacy and security assessment
Editorial and accessibility reviewers	Independent QA gates before release

Independence note (Recommendation). Quality raters should not operate the systems they score, and at least one reviewer external to the drafting team should assess the flagship before release, consistent with the ERC implementation priorities.

Glossary and acronyms

API — hosted model interface billed per token or request. **Break-even point** — utilization or volume where two cost models intersect. **Concurrency** — simultaneous in-flight requests. **Edge device** — embedded or mobile hardware at the point of use. **MLPerf** — MLCommons inference benchmark suite with fixed rules and accuracy targets. **NPU** — neural processing unit. **PUE** — power usage effectiveness; the datacenter overhead multiplier on IT power. **Quantization** — reduced-precision weights used to fit memory and energy budgets. **RAG** — retrieval-augmented generation. **Sustained load** — continuous operation long enough to reach thermal equilibrium. **TCO** — total cost of ownership. **TEVV** — test, evaluation, verification, and validation. **TTFT** — time to first token. **Utilization** — fraction of paid capacity actually used. **ZDR** — zero data retention, a contractual arrangement. **Wilson interval** — a confidence interval for proportions with better small-sample behavior than the normal approximation.

Change log

Table 20. Version history

Version	Date	Change
1.0	2026-10-27	Initial public draft: evidence audit of deployment trade-offs; preregistered workload-based benchmark design; decision matrices and break-even method; timestamped pricing snapshots.
1.0a	2026-10-28	Production remediation: embedded TrueType typefaces; document language (en-US) declared in the PDF catalog; accessibility pre-test record published; companion accessible HTML edition generated.
2.0 (planned)	2027 (post-fieldwork)	Measured results across five deployment models, seven workloads, and ten dimensions; machine-readable results file and calculation workbook; remediated tagged PDF/UA edition.

Data and code statement

This edition contains no ERC-generated datasets; quantitative content is synthesized from cited studies with per-value provenance and timestamped pricing. For Edition 2: the workload definitions, benchmark scripts, raw logs, machine-readable results file (CSV/JSON), and calculation workbook (XLSX) will be published on the ERC canonical report page for

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Corrections. Corrections, replacement versions, the results file, and the calculation workbook will be posted on the canonical ERC report page identified in the document-control section. Readers who identify an error may contact ERC through that page.